Hitesh Rasineni

hiteshrasineni.07@gmail.com • +91 7901076965 • ORCID • LinkedIn • GitHub • The Unlikely Lab

ABOUT ME

CS undergraduate working at the intersection of Machine Learning, statistics, and High Energy Physics, with experience in density estimation and clustering for particle-physics data. Focused on advancing density estimation techniques and applying them to Beyond the Standard Model (BSM) physics.

EDUCATION

VIT-AP University, Amaravati, Andhra Pradesh, India — **September 2023 – September 2027**
B.Tech, CSE (AI & ML) — CGPA: 8.05 / 10

Sree Vidyanikethan International School, Tirupati, India — **2023**
CBSE (10+2) — Grade: 91.6 / 100

SELECTED PROJECTS

The Unlikely Lab — Personal Research Publication Platform hiteshrasineni.github.io/the-unlikely-lab
Code: github.com/HiteshRasineni/the-unlikely-lab

- Designed and deployed a full research-publication platform that hosts complete, typeset HTML versions of arXiv-style physics papers (sections, numbered equations, figures, tables, bibliography) alongside an MDX-driven content system for research notes and project pages.
- Built an automated LaTeX-to-HTML paper build pipeline with Pandoc, including MathML/KaTeX rendering, figure/table validation reports, and per-paper static generation; the site is a statically exported Next.js (React, TypeScript, Tailwind CSS) application deployed via GitHub Actions CI/CD to GitHub Pages.

Cloud-Native ML Training Platform — *FastAPI, Docker, Next.js, PyTorch*
Code: github.com/HiteshRasineni/Cloud-Native-ML-Training-Platform-for-Scientific-Data

- Built a cloud-native ML experiment-orchestration platform for HEP data with a working end-to-end vertical slice: a FastAPI backend (PostgreSQL/Alembic + Redis) validates specs, resolves dataset references against a registry, and enqueues jobs; the scheduler runs a QUEUED → SCHEDULING → RUNNING → COMPLETED/FAILED lifecycle (with retries) via a pluggable executor (Docker now, Kubernetes stubbed), launching workload containers that report per-epoch metrics to MLflow + Redis; the Next.js/TypeScript frontend submits experiments and renders loss curves through a backend MLflow proxy.
- Separated storage responsibilities so PostgreSQL owns lifecycle/scheduling state, the dataset registry and artifact index, MLflow (SQLite) owns params, metrics and model artifacts, and MinIO owns datasets, checkpoints and logs; a pluggable worker workload registry runs a from-scratch RealNVP-style normalizing flow in PyTorch (architecture fully spec-driven) alongside a no-op workload, all orchestrated by a docker-compose stack with pytest suites in the backend and worker.

RESEARCH EXPERIENCE

Leptonic Mono-Z Dark Matter Search with Neural Spline Flows — *CMS Run 2015D Open Data*
Code: github.com/HiteshRasineni/Leptonic-Mono-z-CMS2015-DarkMatter-Search

- Trained five Neural Spline Flows on 2.32 fb^{-1} of CMS Run 2015D data (parallel $\mu\mu/ee$ channels; 40 observables from MINIAOD reduced to a 37-dim feature vector): two channel-specific SM background flows learned from control-region Drell–Yan events, and three mediator-specific signal flows (vector, axial-vector, scalar) from MadGraph-based MC.
- Used the per-event log-likelihood ratio $\log p(x|\text{DM}) - \log p(x|\text{SM})$ as the test statistic, followed by a simultaneous SR+VR binned profile-likelihood fit; set 95% CL observed limits on the signal strength of $\mu < 0.018$ (scalar), 0.036 (vector), and 0.050 (axial-vector), attributing the excess over expectation to a high-E_T^{miss} background-modelling residual rather than a DM signal.

Hadronic Mono-Z Dark Matter Sensitivity with Flow Matching *CMS Run 2015D HTMHT Open Data*
Code: github.com/HiteshRasineni/CMS2015DarkMatterSearch-HTMHT-

- Built the full simulation-to-inference pipeline (MadGraph5_aMC@NLO → Pythia 8 → Delphes signal generation; feature extraction from 20.7M raw events, 1.44M selected across 5 unprescaled HLT paths) and modeled the background density with a conditional flow-matching normalizing flow trained on real HTMHT events.
- Applied sentinel imputation for missing-object features, held-out NLL scoring, an offline signal trigger proxy, and a $B \geq 20$ yield floor to suppress low-statistics argmax bias; projected expected significances of 2.89σ, 7.62σ, and 7.41σ for three simplified-model benchmarks, with ablations showing extra-jet kinematics carry 53–71% of the discriminating power.

Improving Discovery-Significance Stability in Higgs Event Classification *HiggsML $H \to \tau^+\tau^-$ benchmark*
with J.J. Pujari, P.A. Immadi, T. Bikku, R.S. Puppali (VIT-AP University / Amrita).

- Developed a supervised contrastive pre-training framework combined with a parallel FT-Transformer + XGBoost ensemble; repeated 5×5-fold cross-validation showed reduced fold-to-fold variance in Approximate Median Significance (AMS = 3.74 on the full dataset) versus focal-loss training, enabling more stable threshold selection. Published in *Discover Artificial Intelligence* (Springer Nature).

PUBLICATIONS

Dark Matter Mono-Z Analysis

- **Mono-Z Dark Matter Search with Neural Spline Flows Using CMS Run 2015D Open Data.**
  H. Rasineni, B. Chebrolu. arXiv preprint, 2026 (submitted for journal publication). *Leptonic decay channel.*
  DOI: 10.48550/arXiv.2607.13771
- **Hadronic Mono-Z Dark Matter Sensitivity with Flow Matching on CMS Open Data.**
  H. Rasineni, B. Chebrolu. arXiv preprint, 2026. *Hadronic decay channel.*
  DOI: 10.48550/arXiv.2609.02923

Improving Stability of Discovery Significance in Higgs Boson Event Classification using Contrastive Representation Learning.
J.J. Pujari, P.A. Immadi, H. Rasineni, T. Bikku, R.S. Puppala. *Discover Artificial Intelligence* (Springer Nature), 2026.
DOI: 10.1007/s44163-026-01683-5

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript / TypeScript (React, Next.js)

ML/Data: PyTorch, scikit-learn, OpenCV, Pandas, Matplotlib, Density Estimation (Normalizing Flows, Likelihood Ratios)

Web/Platform: Next.js, React, TypeScript, Tailwind CSS, MDX

Tools/Infra: Git, GitHub Actions (CI/CD), Docker, Pandoc, LaTeX, uproot, Google Colab

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals** (Credential)
- **AWS Academy Graduate – Cloud Foundations (Training Badge)** (Credential)
- **WorldQuant University, Applied AI Lab – Deep Learning for Computer Vision** (YOLOv8, MTCNN, Inception-ResNetV1) (Credential)
- **IBM Machine Learning Specialist (Professional V1)** (EDA, ensembles, transfer learning) (Credential)
- **NVIDIA DLI – Fundamentals of Deep Learning** (Credential)
- **NVIDIA DLI – Transformer-based NLP** (Credential)